

Quantitative Susceptibility-Mapping Reconstruction

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16.1 Introduction

Quantitative Susceptibility Mapping (QSM) aims to estimate tissue magnetic susceptibility distribution, χ_{tissue} , that gives rise to subtle changes in the main magnetic field of an MRI scanner. These effects can be in the range of ∓ 10 Hz and thus remain in the parts-per-million (ppm) level relative to the Larmor frequency. Despite being small, these changes in the magnetic field can provide a significant contrast to the noise ratio (CNR) boost over standard GRE magnitude images when certain post-processing steps are applied [1] (Fig. 16.1, top). Unfortunately, this tissue field map, δ_{tissue} , depends on the orientation of the tissue relative to the main magnetic field. This implies that, if the same subject is imaged at different positions, drastically different tissue field-map information may be obtained. Fig. 16.1 (bottom) shows the effect of this in imaging of the brain, the most commonly targeted organ in QSM, where subjects were asked to tilt their head to multiple positions.

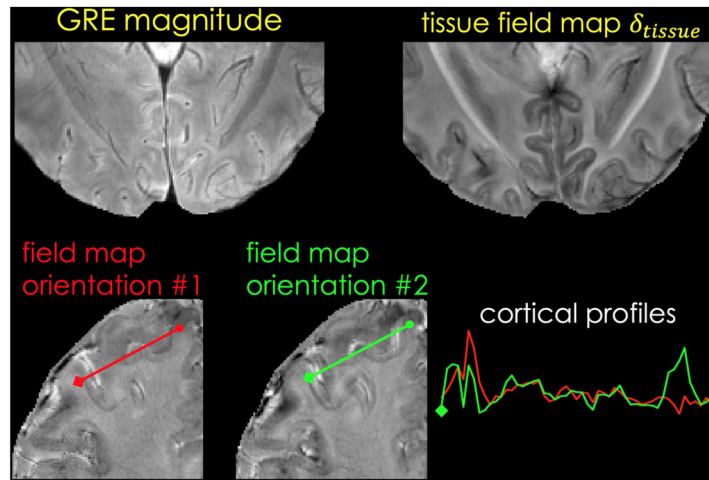
Another complication in the interpretation of tissue field maps is the influence of a spatial “blur” imposed by the dipole kernel d (Fig. 16.2). This additional blur stems from dipole physics [2], and can be expressed as

$$\chi_{\text{tissue}} \otimes d = \delta_{\text{tissue}}. \quad (16.1)$$

Accordingly, there is a convolutional relationship between the unknown susceptibility map χ_{tissue} and the acquired field map. This convolution, denoted as $\otimes$, causes a mismatch between the tissue field map and the underlying anatomical boundaries of tissues. QSM aims to deconvolve this dipole kernel to eliminate the orientation bias and spatial blurring effects, while providing a quantitative and sensitive biomarker. The convolutional relationship can be conveniently expressed in k-space via multiplication with the frequency-domain representation of the dipole kernel, D , as follows

$$F^{-1} D F \chi_{\text{tissue}} = \delta_{\text{tissue}}, \quad (16.2)$$

where F denotes 3D discrete Fourier transform.

**FIGURE 16.1**

top: Tissue field-map provides a significant CNR boost of the GRE magnitude signal, especially at ultra-high fields. **bottom:** However, this field map is dependent on the orientation of the head relative to the main magnetic field, which can result in large differences when plotted as a 1D signal profile in the cortex, despite coregistering the two acquisitions.

FIGURE 16.2

Tissue susceptibility is related to the acquired field map through a convolution, which can be represented with a voxel-wise multiplication in frequency space (Eq. (16.2)). This convolution operation creates a spatial blur, which complicates the interpretation of the field map since this information does not accurately depict the underlying anatomical boundaries. This can best be visualized in the basal ganglia, as indicated by the yellow (light gray in print version) oval.

This imaging biomarker has been found to be highly correlated with tissue iron concentration, especially in the basal ganglia [3], and has been used to estimate oxygen concentration in the brain vessels [4]. QSM also provides high enough CNR to provide exquisite details at mesoscale spatial resolutions, consistent with the underlying histology [5]. It is important to note that neuroimaging has been the predominant focus of QSM, and, as such, proposed reconstruction techniques have been specialized to the brain. However, recent developments have expanded the application domain of QSM to abdominal [6], cardiac [7], bone [8], and breast imaging [9]. Imaging these organs has been hampered by chal-

lenges, including the presence of fat signal, motion, and, in the case of bone imaging, rapid T_2^* decay. These challenges have been mitigated by advanced water–fat signal modeling, breath-held and gated acquisitions, and ultra-short echo times, respectively.

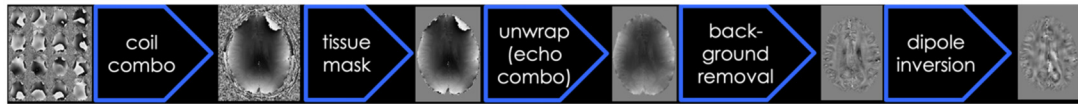
An additional challenge is that, the dipole inversion problem (Eq. (16.1)) is an ill-posed and ill-conditioned linear inverse problem. The equation is not well satisfied due to physical and physiological biases in the data. Moreover, the kernel D undersamples the frequency content of the susceptibility map and is not directly invertible. Before tackling this inverse problem, we first detail data-acquisition aspects and provide a checklist of desirable choices so that the acquired GRE volume lends itself well to reconstruction by the methods presented in the following. We also describe the pre-processing steps that need to be applied on the acquired GRE signal phase to obtain the desired tissue field map. We define and discuss the implementation of these pre-processing steps, detail popular algorithms to solve the inverse problem, and focus on “single-step” approaches that merge the pre-processing and dipole inversion steps. We conclude the chapter by reporting about current challenges, as well as some of the future directions in the field.

16.2 GRE data acquisition

The same GRE acquisition lends itself to both QSM and susceptibility weighted imaging (SWI) [10] processing. Clinical SWI brain-imaging acquisitions are mostly made with a 3D-GRE sequence using high in-plane and relatively low-slice resolution, e.g., $1 \times 1 \times 2 \text{ mm}^3$. Since a minimum intensity projection (MIP) across the slices is usually performed, such anisotropic voxel sizes can be conveniently used in SWI. For QSM, however, the 3D nature of the dipole convolution often calls for isotropic voxel sizes. A common choice is 1-mm isotropic resolution, which usually has ample SNR at 3T. Although 2D-GRE acquisitions lend themselves to QSM processing as well, they usually fail to match the SNR gain provided by the volumetric noise-averaging ability of 3D-GRE sequences.

A constraint in 3D-GRE acquisitions that may dictate the achievable resolution is the scan time. The magnitude signal in GRE scans decays with T_2^* as a function of the echo time (TE). As such, shorter TEs provide increased magnitude SNR. This is convenient since a short TE would enable a short repetition time (TR) and reduce the overall scan time. The phase SNR, however, is maximized when TE equals to the T_2^* of the tissue of interest [11]. This would require an optimal TE of, e.g., 35 msec at 3T for the brain parenchyma, thus pushing the TR to around 40 msec. Assuming matrix sizes of 240 in phase- and 120 in partition-encoding directions for 1-mm³ resolution with whole-brain coverage, this would necessitate a ~19-min scan without acceleration. In QSM (and SWI), it is desired to maximize the phase SNR since this is the input data used in subsequent processing. Reaching resolutions such as 1-mm isotropic within clinically feasible scan times of, e.g., 5 min for QSM acquisition, would require additional acceleration. With standard parallel-imaging methods such as SENSE [12] or GRAPPA [13], achieving $R = 2 \times 2$ -fold acceleration with modern head coils (e.g., 32 channel) is possible and would render such 5-min QSM acquisition feasible.

Having identified a 5-min 3D-GRE brain-imaging acquisition at 1-mm isotropic resolution at TE/TR = 35–40 msec as a desirable target at 3T, we can next consider boosting its sampling efficiency. One trivial option would be to sample a single echo, and use as low of a readout bandwidth as possible to improve the SNR of this single volume. This would still leave significant unused sequence time prior to this single readout, which can be better utilized by sampling additional, earlier echoes.

**FIGURE 16.3**

Phase pre-processing steps include coil combination, tissue masking, phase unwrapping (and echo combination in multi-echo acquisitions), and background phase removal. At the completion of these steps, the obtained tissue field map is used in the final dipole-inversion step to estimate the susceptibility map.

Having multiple echoes would both provide the ability to quantify T_2^* maps as an additional quantitative contrast, and also yield an SNR boost via combining the phase information across the echoes. We will touch upon multi-echo combinations in the following section. Finally, we recommend using unipolar echoes to simplify phase combination across the echoes. Although bipolar sampling provides higher efficiency, subsequent echoes suffer from geometric distortions in opposite directions along the readout direction, which need to be corrected before echo combination. The final protocol we recommend is a 5-min, unipolar multi-echo 3D-GRE acquisition using $R = 2 \times 2$ parallel imaging acceleration at $TR = 40$ msec and sampling multiple echoes ranging from early TEs to the ones closer to the optimal $TE = T_2^*$ point.

16.3 Phase pre-processing

Phase pre-processing pipeline comprises coil combination, tissue masking, phase unwrapping, and background field removal (Fig. 16.3). Each of these steps is discussed hereafter; example links to openly available code are provided.

3.1 Coil combination: is a crucial first step since potential errors may propagate all the way to the final susceptibility map. These errors stem from imperfect estimation and removal of spatially varying phase offsets of each receiver coil. Estimating the coil sensitivities makes it possible to remove the spatially varying coil-phase offsets and obtain an SNR-optimal coil combination. This is referred to as Roemer [14] or SENSE combination [12] and is usually implemented in standard GRE acquisitions. If this is not present in the vendor-provided sequences, either a separate low-resolution reference scan or the fully sampled autocalibration signal (ACS) region of the GRE acquisition can be used in coil sensitivity estimation. Further details can be found in Chapter 6. ESPIRiT algorithm [15] allows for automated sensitivity estimation and is freely available as a part of the BART toolbox (<https://mrrecon.github.io/bart/>). ESPIRiT provides *relative* coil-phase offsets, and thus needs a coil that can act as the phase reference. This reference coil needs to be free of phase singularities and can be obtained using body coil reception, or with virtual coil computation [16]. At ultra-high fields, there is usually no body coil that would serve as a reference coil, and the simple virtual coil approach may include phase singularities since the sensitivity profiles vary more rapidly in space. To combat this, more involved virtual reference-coil computations, such as block coil compression (BCC), can be used [17] (available at <https://bit.ly/370SrI8>).

While using an additional GRE calibration scan allows for coil sensitivity mapping, it can also be directly utilized for coil combination without the need for explicit sensitivity estimation. COMPOSER uses a low-resolution, short-TE acquisition for this purpose [18]. This models the acquired coil-phase signal ϕ_{acq} as the summation of a coil phase offset ϕ_{coil} and a time-dependent term:

$$\phi_{acq} = \phi_{coil} + 2\pi \cdot TE \cdot f \quad (16.3)$$

where f denotes the frequency map and TE is the echo time. It can then be seen that using a separate calibration acquisition with $TE \approx 0$ gives a good approximation to the underlying coil-phase offsets due to $\phi_{acq} \approx \phi_{coil}$. Another multi-echo acquisition/reconstruction approach is ASPIRE [19], which is publicly available (<http://github.com/korbinian90/ASPIRE>) and permits online coil combination at the scanner console.

3.2 Masking: This step becomes necessary for most of the algorithms that are used in the subsequent pre-processing steps. Limiting the computation inside a mask helps improve the processing speed of most phase-unwrapping techniques. During background field removal, large phase contributions due to air–tissue interfaces are estimated and removed. This necessitates masking so that tissue and background regions of interests (ROIs) can be defined. Dipole inversion step that estimates the susceptibility distribution from the tissue phase may also benefit from masking, or from conditioning the inverse problem using an organ-masked weighting matrix.

Current open-source tissue masking algorithms focus on brain imaging. FSL’s BET [20] is by far the most commonly used tool for brain imaging (<https://fsl.fmrib.ox.ac.uk>). It is fast (on the order of seconds) and robust, albeit in ultra-high fields it may benefit from tuning the parameter that controls the mask size. The power of deep learning in segmentation and classification tasks has also been leveraged for masking. An open-source tool for brain imaging is *deepbrain* (<https://github.com/iitcco/deepbrain>), which aims to improve the robustness and speed of BET.

3.3 Phase unwrapping: Having combined coils (and potentially performed masking), ambiguities caused by multiples of 2π phase jumps need to be removed from the phase image. This process is called phase unwrapping and can be carried out in numerous different ways. Software tools such as FSL’s PRELUDE [21], as well as computationally faster SEGUE [22] (<https://xip.ucsb.edu/i/software/SEGUE.html?print>) and ROMEO [23] (<https://github.com/korbinian90/ROMEO>) provide open-source solutions to this problem. An alternative and robust technique is Laplacian unwrapping [24,25]. This can be carried out with or without tissue masking using the following relationship:

$$\Psi(\phi) = \Delta^{-1} \cdot \text{Imag} \left(e^{-j\phi} \cdot \Delta e^{j\phi} \right) = \Delta^{-1} \cdot (\cos\phi \cdot \Delta \sin\phi - \sin\phi \cdot \Delta \cos\phi), \quad (16.4)$$

where $\Psi(\phi)$ is the unwrapped phase, Δ is the 3D Laplacian operator, and $\text{Imag}()$ denotes the imaginary component. Source code and exemplar data that implement Eq. (16.4) are included in *Tutorial 16.A.1*, and are also a part of the STI Suite (<https://people.eecs.berkeley.edu/~chunlei.liu/software.html>). It should be noted that Laplacian unwrapping is not an unwrapping operator in the sense that the residual $(\Psi(\phi) - \phi)$ is not guaranteed to be an integer multiple of 2π . This, however, may not be crucial because additional background filtering needs to be applied after phase unwrapping, and conventional and Laplacian unwrapping techniques yield very similar results when these filtered tissue

field maps are compared [26]. Laplacian unwrapping has emerged as a popular choice since it can relate the wrapped phase to the unwrapped phase through simple trigonometric functions and it readily eliminates some background signals as they satisfy the Laplace equation ($\Delta\phi = 0$).

3.4 Echo combination: In multi-echo acquisitions, a separate image volume is obtained per each echo time. As such, phase information from these multiple images needs to be combined into a single-phase image. One simple way of achieving this is to sum the unwrapped phase from each echo. Since phase unwrapping is not a linear operator, this needs to be performed before summing the echo phase images. The echo-combined phase image ϕ_{sum} is given by

$$\phi_{sum} = \sum_t \phi_t = 2\pi \cdot f \cdot \sum_t T E_t, \quad (16.5)$$

where ϕ_t is the unwrapped phase image from the t^{th} echo and $T E_t$ is the corresponding echo time. Assuming that phase evolves linearly with time and the underlying frequency map is denoted by f , it can be seen that the “effective TE” of this echo-combined phase image is $\sum_t T E_t$. This allows us to compute a frequency map f_{sum} normalized by the effective TE due to $f_{sum} = \frac{\phi_{sum}}{2\pi \cdot \sum_t T E_t}$.

An alternative approach is to normalize the echo-phase images by their TE, and then perform the averaging

$$f_{avg} = \sum_t \frac{\phi_t}{2\pi \cdot T E_t} \quad (16.6)$$

This yields an average frequency map f_{avg} . The main difference between Eqs. (16.5) and (16.6) is the SNR weighting: Eq. (16.5) weights the frequency map f of each echo by its TE, whereas Eq. (16.6) normalizes the effect of TE before summing. Since early echoes have low-phase SNR, it is beneficial to down-weight their contribution. Eq. (16.5) aims to achieve this using TE-weighting, but this may become suboptimal at the very late TEs since the magnitude signal will decay with T_2^* and become negligibly small. A better approach could be using an SNR-optimal combination of echoes as per [11]:

$$f_{weighted} = \sum_t \frac{\phi_t}{2\pi \cdot T E_t} \cdot w_t, \quad (16.7)$$

where the weighting amount of the t^{th} echo w_t is equal to $\frac{T E_t \cdot e^{-T E_t / T_2^*}}{\sum_t T E_t \cdot e^{-T E_t / T_2^*}}$. The T_2^* map that is required in this equation can be readily estimated, e.g., using nonnegative least-squares fitting to the echo-magnitude images. Eqs. (16.5)–(16.7) can be applied after background-field removal as well, but, since background filtering is a linear operation, it is computationally more efficient to perform echo combination first.

Another echo-combination technique is applied on the complex-valued signals directly [27]. This technique estimates a frequency map and a phase offset that match the acquired complex multi-echo images by fitting a complex exponential across the echoes; it is publicly available in the MEDI toolbox (<http://pre.weill.cornell.edu/mri/pages/qsm.html>). Compared to the preceding echo-combination techniques, this exponential fit is applied prior to phase unwrapping. This estimated frequency map can contain phase wraps and will benefit from unwrapping.

3.5 Background field removal: So far, we have combined coils, performed phase unwrapping, and, for multi-echo acquisitions, combined echoes as well. Using either echo summation, averaging, or weighted combination, these steps yield the frequency map f_{total} . This can be further normalized by the gyromagnetic ratio γ and the field strength B_0 to switch to a unitless scale where the influence of field strength can be eliminated. This normalized field map $\delta_{total} = f_{total}/(\gamma \cdot B_0)$ is expressed in parts per million (ppm), and is approximately the summation of a foreground tissue component δ_{tissue} and an unwanted background signal δ_{back} .

The background-field map δ_{back} stems from several sources including large susceptibility differences between air–tissue and bone–tissue interfaces, the static main magnetic field inhomogeneity, and macroscopic currents [28], and can be more than an order of magnitude stronger than the tissue field map of interest. As such, estimating and eliminating this background component is critical.

Based on underlying assumptions, problem formulation, and characteristics of the solution, background-field removal methods can be rigorously categorized into three types [29]: assumption of no sources close to the boundary (NOS), assumption of no harmonic (NOHA) internal and boundary fields in the boundary region, and minimization of an objective function involving a norm (MOIN). Each type has its own advantages and limitations (summarized in Table 1 of [29]), and, hence, one type may be preferred to the others depending on the situation. According to quantitative comparisons of different background-field removal methods on a numerical model, all the methods performed similarly in deep brain regions, whereas NOS methods resulted in less reliable foreground-tissue component (i.e., internal fields) δ_{tissue} in the cortical regions of the brain [29]. In this chapter, we provide an overview of two of the most widely used background field-removal methods that have repeatedly demonstrated their robustness on *in vivo* data: projection onto dipole fields (PDF) [30] and sophisticated harmonic-artifact reduction for phase data (SHARP) [31].

PDF, a NOHA method, aims to place a background susceptibility distribution χ_{back} outside of the brain that best explains the total field map δ_{total} inside the tissue mask M :

$$\chi_{back}^{pdf} = \operatorname{argmin}_{\chi_{back}} \|M(F^{-1}DF\underline{M}\chi_{back} - \delta_{total})\|_2^2 \quad (16.8)$$

where $\underline{M}$ denotes the complement of the tissue mask (i.e., background mask). Having found this background susceptibility distribution, its effect inside of the tissue foreground is computed by convolving it with the dipole kernel, $\delta_{back}^{pdf} = MF^{-1}DF\underline{M}\chi_{back}^{pdf}$. This estimated background-field map can then be subtracted from the total field to yield the tissue field map estimate due to $\delta_{tissue}^{pdf} = \delta_{total} - \delta_{back}^{pdf}$. An implementation of PDF is included in the MEDI Toolbox.

On the other hand, SHARP, a NOS method, relies on the assumption that the background-field map is harmonic inside the mask. This allows us to invoke the spherical mean-value (SMV) property of harmonic functions, which states that a harmonic function is preserved when convolved with a radially symmetric, nonnegative kernel that sums up to one. As such, when a radial filter r is applied on the total field map, it should eliminate the background component

$$\delta_{total} - r \otimes \delta_{total} = \delta_{back} - r \otimes \delta_{back} + \delta_{tissue} - r \otimes \delta_{tissue} \approx (u - r) \otimes \delta_{tissue} \quad (16.9)$$

where u denotes the Dirac delta function, and we used $\delta_{back} \approx r \otimes \delta_{back}$. We define $h = (u - r)$ to be an SMV filter, which can be applied rapidly using multiplication with its k-space counterpart H . An issue with SHARP arises at the boundary of the tissue foreground mask, where convolution with the

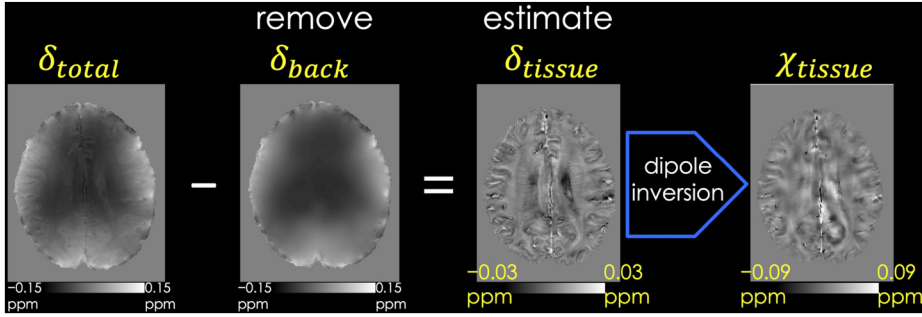


FIGURE 16.4

Background-field removal aims to eliminate contributions from background susceptibility sources that induce slowly varying fields inside of the tissue mask, by which the tissue field map is obtained. This is followed by the dipole inversion step that estimates the tissue susceptibility that gave rise to the tissue field map. Note that the dynamic range of the tissue field map is 5x smaller than those of the total and background fields.

radial filter has contributions from both the tissue and background components due to the extent of the kernel. This ambiguity necessitates eroding several voxels away from the surface of the brain mask and leads to the following linear system:

$$M_{sharp} F^{-1} H F \delta_{total} = M_{sharp} F^{-1} H F \delta_{tissue}, \quad (16.10)$$

where M_{sharp} is the eroded mask. Eq. (16.10) is underdetermined, and its conditioning can be improved using an SMV with a larger radius. However, a larger filter would require more erosion and further reduce the size of the tissue mask. To address this complication, V-SHARP proposes to use multiple filters with varying radii [32]. Large filters provide high-fidelity tissue field-map estimates in the interior of the object, whereas using smaller filter sizes towards the edge of the tissue helps preserve more information at the tissue surface. V-SHARP filtering is governed by the following equation,

$$\sum_i M_i F^{-1} H_i F \delta_{total} = \sum_i M_i F^{-1} H_i F \delta_{tissue} \quad (16.11)$$

where M_i represents the reliable field-map mask belonging to the i^{th} SMV filter H_i . Eq. (16.11) can be solved iteratively with Tikhonov regularization [33], or the impact of SMV filtering can be deconvolved due to

$$\delta_{tissue}^{v-sharp} = F^{-1} H_{inv} F \sum_i M_i F^{-1} H_i F \delta_{total}, \quad (16.12)$$

where H_{inv} denotes the inverse of the largest SMV filter, which needs to be truncated with a certain threshold parameter [34]. This deconvolution step may also be bypassed since large SMV filters have a flat point-spread function in k-space [32]. An implementation of V-SHARP is included in Tutorial 16.A.1, as well as in the STI Suite. The background field removal step is depicted in Fig. 16.4, which is followed by the dipole inversion step that will be covered in the next section.

16.4 Dipole inversion

In the previous section, several pre-processing steps were introduced to convert the raw GRE phase to a tissue field map. In the last step of the reconstruction, an inverse problem known as the dipole inversion needs to be solved to estimate the desired susceptibility. The QSM forward model relates the susceptibility χ_{tissue} to the tissue field δ_{tissue} using the dipole kernel d . This relation can either be expressed as a convolution in the image domain (Eq. (16.1)) or more conveniently as a multiplication in k-space (Eq. (16.2)). Unfortunately, this inverse problem is ill-conditioned because the conical dipole kernel in k-space has diminishing values on its surface (Fig. 16.2) and thus undersamples the frequency space of the susceptibility χ_{tissue} . This impedes solving Eq. (16.2) using a simple voxel-wise kernel division in k-space because this would cause streaking artifacts in the reconstructed susceptibility map.

The inversion of the QSM forward model does not lead to a unique solution and hence multiple susceptibility distributions are in agreement with the acquired MR phase signal. To identify the physically meaningful solution, one can either aim to “fully sample” the susceptibility by acquiring data in multiple orientations with respect to the main magnetic field B_0 , or utilize regularization in the reconstruction. Regularized approaches can be divided into image- and k-space methods that either allow a closed-form solution or necessitate iterative optimization. In the following, four common dipole inversion techniques are briefly introduced, and example reconstructions are provided in Fig. 16.6.

16.4.1 COSMOS

The dipole inversion in QSM is ill-posed and ill-conditioned, e.g., without prior constraints, an image reconstruction will assign arbitrary values in the zero-cone neighborhood of the dipole kernel in k-space. Calculation of susceptibility using multiple orientation sampling (COSMOS) [35] recovers the “missing” k-space information from these regions. This is achieved by changing the orientation of the subject’s head with respect to the main magnetic field B_0 and reacquiring the MR data. In the COSMOS reconstruction, the tissue field maps from all orientations are coregistered. Then, an overdetermined linear system of equations is solved to estimate the susceptibility map that explains the tissue field-map data from each orientation.

Using the head orientation index $j \in \{1, \dots, N\}$, where D_j corresponds to the dipole kernel for the j^{th} orientation in k-space, this inverse problem can be formulated in the matrix form

$$\begin{bmatrix} D_1 \\ \vdots \\ D_N \end{bmatrix} F\chi = \begin{bmatrix} F\delta_1 \\ \vdots \\ F\delta_N \end{bmatrix}. \quad (16.13)$$

It also admits the closed-form solution $\chi = F^{-1}(\sum_j D_j^T D_j)^{-1} \sum_j D_j^T F\delta_j$ and thus allows reconstructions with minimal computational footprint.

Reorientation and reacquisition of MR data at different angles with respect to the main magnetic field reduces the ambiguity of the ill-conditioned dipole inversion. This is because the dipole kernel rotates for each orientation and, hence, spatial frequency undersampling due to diminishing values on the surface of the conical dipole kernel in k-space vary in position (Fig. 16.5). This facilitates QSM reconstruction as the dipole inversion problem is no longer underdetermined. Unfortunately, the need for multi-orientation data also leads to longer scan times which have impeded widespread clinical

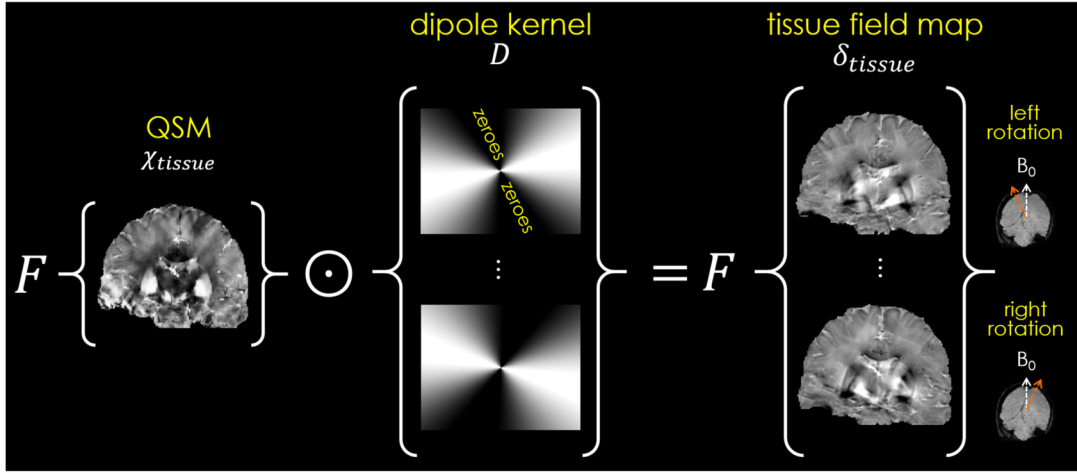


FIGURE 16.5

COSMOS recovers “missing” k-space information in the zero-cone region of the dipole kernel by re-orienting the subject with respect to the main magnetic field B_0 and resampling the MR data. After coregistration of the tissue field maps, an overdetermined linear system of equations is solved. COSMOS reduces the ambiguity of the dipole inversion since the position of the zero-cone area of the dipole kernel varies in each orientation, thus each tissue field map provides complementary k-space information about the desired susceptibility.

usage. However, COSMOS often serves as a ground-truth reference when evaluating and comparing dipole-inversion techniques.

16.4.2 K-space reconstruction with closed-form solution

In the previous section, the conditioning of the dipole inversion was improved by better sampling the susceptibility distribution through the acquisition of multi-orientation data. In this section, regularization is employed to identify a unique solution. Regularization can be applied directly in k-space, and truncated k-space division (TKD) [36] is the most commonly used technique among the k-space methods. It inverts the dipole kernel D directly and replaces small absolute values in D by a constant number ϵ .

$$\hat{D}(k) = \frac{\text{sign}(D(k))}{\max(|D(k)|, \epsilon)} \quad (16.14)$$

Here, the division is performed in an element-wise fashion, and $D(k)$ denotes the k^{th} diagonal element of the matrix D . The desired susceptibility is then obtained using the closed-form solution that involves Fourier transforms and simple element-wise matrix multiplication, $\chi = F^{-1} \hat{D} F \delta_{\text{tissue}}$. Note that closed-form solutions also exist for, e.g., quadratic (L2) gradient penalties [37]. Unfortunately, the modification of the dipole kernel in k-space can lead to systematic underestimation of the tissue susceptibility in k-space, especially around the zero-cone region of the dipole kernel. Corresponding gaps

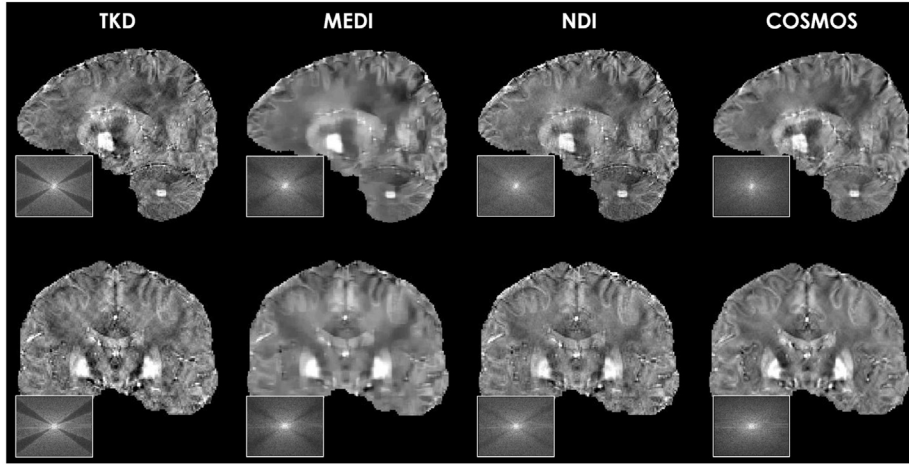


FIGURE 16.6

Comparison of single-orientation TKD, NDI, and MEDI. COSMOS was computed on data from five head orientations and serves as a ground-truth reference. The truncation of the dipole kernel in TKD results in residual streaking artifacts and underestimation of the susceptibility signal in image space. In k-space these artifacts manifest as gaps around the zero-cone of the dipole kernel. These artifacts can be mitigated, albeit at the cost of oversmoothing in MEDI. The gradient-based regularization suppresses streaking artifacts and fills gaps in k-space. Similar image quality is also obtained from NDI, which does not use explicit regularization but relies on the inherent regularization effect from magnitude weighting and nonlinear formulation.

in k-space typically manifest as streaking artifacts and noise amplification in the image domain [38], which is also seen in the example reconstruction in Fig. 16.6.

16.4.3 Iterative reconstructions in image space

Several dipole inversion techniques that exploit prior constraints in the image domain have been proposed. These priors are applied either to the image itself or a representation using a transform. The image reconstruction then involves the minimization of a cost function that typically follows Eq. (16.15).

$$\text{minimize}_{\chi} \frac{1}{2} \left\| F^{-1} D F \chi - \delta_{\text{tissue}} \right\|_2^2 + \lambda R(\chi) \quad (16.15)$$

The first term is the data-consistency term which enforces agreement with the QSM forward model. The second term is the regularizer that imposes prior constraints on the reconstructed susceptibility map (such as image smoothness). A regularization parameter λ is used to balance data-consistency (model accuracy) and regularization error and needs to be tuned for a specific application.

In the literature, a wide range of regularized dipole-inversion techniques has been reported, which mainly differ in the choice of the regularizer [27,37,39–41]. While the majority of these methods require iterative optimization, some of them also admit closed-form k-space solutions. In this section, we will briefly introduce Morphology-Enabled Dipole Inversion (MEDI) [39], a commonly used iterative

technique. In MEDI, anatomical information from the magnitude image is used as a prior for the image reconstruction. This is based on the assumption that edges in the desired susceptibility should be colocated with anatomical edges in the known GRE magnitude image. MEDI (Eq. (16.16)) incorporates this idea into the regularization term where a 3D gradient operator ∇ acts on the susceptibility χ . The gradient is weighted by a diagonal binary matrix M , derived from the magnitude image, to exclude known anatomical edge structures from the gradient penalty. The use of the L1-norm then enforces gradient sparsity and thus suppresses streaking artifacts. Moreover, a diagonal noise-weighting matrix W is included in the data-consistency term to account for the nonuniform phase-noise distribution across the image.

$$\underset{\chi}{\text{minimize}} \frac{1}{2} \left\| W(F^{-1}DF\chi - \delta_{\text{tissue}}) \right\|_2^2 + \lambda \|M\nabla\chi\|_1 \quad (16.16)$$

In practice, the acquired MR data are contaminated by noise, which can lead to unreliable field map values. Especially in low-SNR regions, the phase-noise distribution can deviate from a Gaussian distribution and is hence not governed by the linear susceptibility-to-field relationship shown in Eq. (16.15). This mismatch can lead to artifacts in the reconstructed susceptibility map, particularly in patients who exhibit strong susceptibility signals, e.g., due to hemorrhages. This issue was recognized in the nonlinear-MEDI (NMEDI) approach [27], which employs a nonlinear data-consistency term (Eq. (16.17)) to better model the complex MR signal and noise distribution.

$$\underset{\chi}{\text{minimize}} \frac{1}{2} \left\| W(e^{iF^{-1}DF\chi} - e^{i\delta_{\text{tissue}}}) \right\|_2^2 + \lambda \|M\nabla\chi\|_1 \quad (16.17)$$

Regularized reconstructions, including (N)MEDI, typically require tuning of the regularization parameter λ . This value trades-off model accuracy and image smoothness, and the optimal choice of this parameter can be subject- and application-specific. Moreover, nonlinear regularizers, as used in (N)MEDI, often require complicated iterative optimization techniques that can be time-consuming. The recently proposed FANSI algorithm [42] aims to address this issue. It minimizes the same cost function as in NMEDI (Eq. (16.17)) but relies on a different optimization strategy (Split Bregman method [43] and ADMM [44]), which transforms the regularization penalty and nonlinear data consistency into simpler decoupled problems that can be solved more rapidly. In particular, parameter splitting allows for using a voxel-wise Newton–Raphson approach to rapidly solve for the roots of the nonlinear data-consistency function with closed-form iterations. While this can provide up to ten-fold speedup, it comes at the cost of additional regularization parameters that need to be tuned. Implementations of both (N)MEDI and FANSI are publicly available (<http://pre.weill.cornell.edu/mri/pages/qsm.html>; <https://gitlab.com/cmilovic/FANSI-toolbox>).

So far, we have briefly introduced popular dipole-inversion techniques that impose prior constraints by explicitly regularizing the reconstructed susceptibility map. In the recently proposed NDI technique (Nonlinear Dipole Inversion, [45]), regularization terms are removed, and the inherent regularization effect from magnitude weighting and nonlinear formulation is exploited. The NDI cost function is a special case of NMEDI and is obtained by choosing $\lambda = 0$ in Eq. (16.17).

$$f(\chi) = \frac{1}{2} \left\| W(e^{iF^{-1}DF\chi} - e^{i\delta_{\text{tissue}}}) \right\|_2^2 \quad (16.18)$$

The removal of gradient-based regularization admits a simple gradient-descent optimization via the analytical derivative $\nabla_{\chi} f(\chi)$ [39]. Using an iterative update rule, the t^{th} image estimate in the NDI

reconstruction becomes

$$\chi^{[t+1]} = \chi^{[t]} - D^T W^T W \sin(F^{-1} D F \chi^{[t]} - \delta_{tissue}) \text{ with } \chi^{[0]} = 0, \quad (16.19)$$

which can be implemented, e.g., in MATLAB®, using just a few lines of code, and is explored in Tutorial 16.A.2.

Gradient-descent optimization (see Chapter 3) is often prone to error propagation/noise amplification in ill-conditioned problems. As previously reported [2], early iterations mainly fill k-space outside of the zero-cone region, while later iterations insert structured noise into this area to further reduce the cost function. Unfortunately, the latter often causes streaking artifacts and noise amplification in the image domain. NDI also suffers from this overfitting problem when too many iterations are performed. However, this issue can be mitigated by the use of early-stopping (implicit regularization) or a small amount of Tikhonov regularization. This improves the convergence of the optimization and typically does not result in oversmoothing, which is often observed in highly regularized techniques.

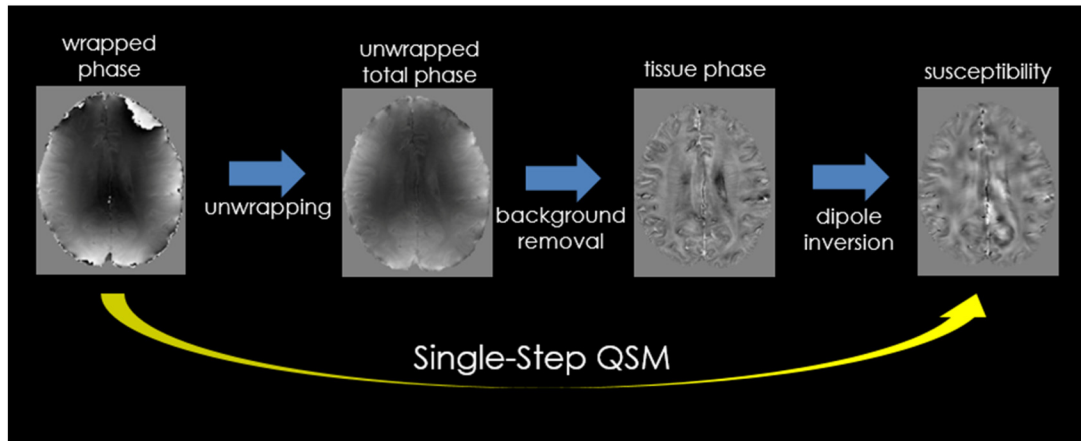
16.5 Recent advances: single-step QSM and deep-learning-based QSM

The QSM methods that we have discussed so far are made up of a sequence of multiple processing steps including phase unwrapping, background-field removal, and dipole inversion. These multistep QSM methods often suffer from error propagation between these processing steps. For instance, an erroneous phase-unwrapping algorithm that gives rise to a corrupted unwrapped phase adversely affects the result obtained from the subsequent background field-removal step. The errors from both the phase unwrapping and background-field removal steps could then get propagated to the dipole-inversion step.

In order to tackle such an error propagation, QSM methods that perform the processing steps simultaneously have been proposed [16,40,46–53] (Fig. 16.7). Several optimization problems, each with the objective function that simultaneously incorporates the phase unwrapping, background-field removal based on the mean-value property of harmonic functions, and regularized dipole inversion components have been formulated [16,40,46–51]. Despite sharing the same principle, these proposed formulations differ in how each of the processing steps is incorporated. They all use Laplacian-based methods to perform phase unwrapping and background-field removal. However, a single Laplacian operator is used in [40,46,48,51], whereas multiple operators are used in [16,47,49,50]. The regularization part of the objective functions also differs: the l_2 -norm of the weighted gradients of the tissue magnetic susceptibility used in [46,47], weighted total variation used in [16,48–50], and total generalized variation used in [16,40,50,51]. Some of these methods have been made publicly available (<http://martinos.org/~berkin/software.html> and <http://qsm.neuroimaging.at>). The general optimization problem solved in single-step methods is of the form

$$\underset{\chi}{\text{minimize}} \frac{1}{2} \left\| \sum_i M_i F^{-1} H_i D F \chi - M_i F^{-1} H_i F \delta_{tissue} \right\|_2^2 + R(\chi), \quad (16.20)$$

where H_i and M_i again denote the i^{th} SMV kernel and reliable mask region and $R(\cdot)$ is a regularizer. The solution to these single-step optimization problems is an estimate of the underlying magnetic susceptibility map in the region of interest, χ_{tissue} . It has been shown that these single-step QSM

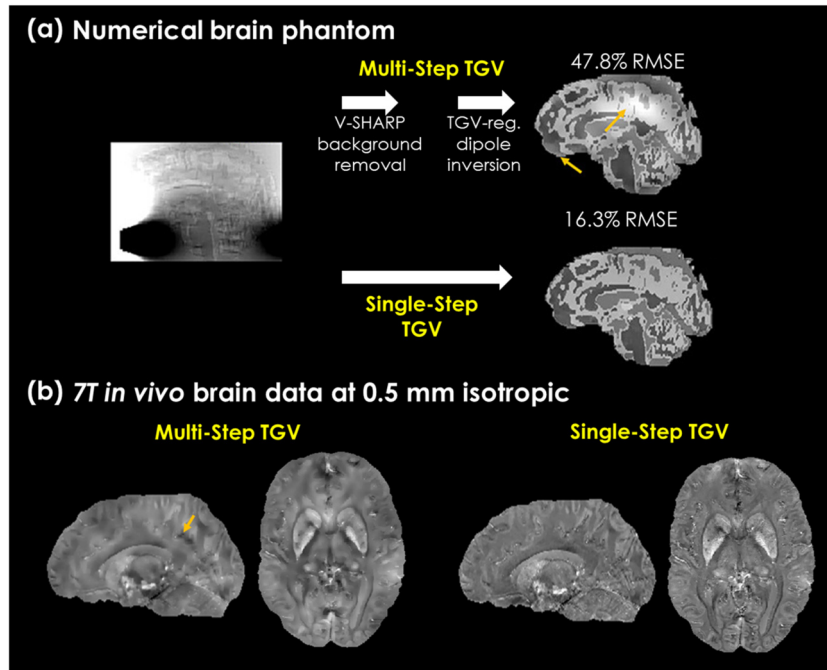
**FIGURE 16.7**

As opposed to a standard multistep QSM pipeline, which performs phase unwrapping, background field removal, and dipole inversion sequentially (the blue (dark gray in print version) arrows), single-step QSM methods perform these operations simultaneously (the yellow (light gray in print version) arrow).

methods resulted in magnetic susceptibility estimates with fewer artifacts than those of the multistep QSM methods [16,40,46–53]. In Tutorial 16.A.3 at the end of this chapter, we will implement the total variation-regularized single-step QSM method proposed in [16,50] and compare it to some of the multistep QSM methods.

The Laplacian-based single-step QSM methods mentioned in the preceding paragraph, especially those that use a single Laplacian operator, typically lose the information near regions with a large susceptibility dynamic range. To address this “erosion” problem, the total field inversion (TFI) [52] and its fast version [53] have been proposed. The TFI methods model the total susceptibility map, χ_{total} , as the summation of the susceptibility map inside a region of interest, χ_{tissue} , and the susceptibility map outside the region of interest, χ_{back} , as $\chi_{total} = \chi_{tissue} + \chi_{back}$. Under this model assumption, χ_{tissue} and χ_{back} can be jointly estimated without having to evaluate the Laplacian, shielding the TFI methods from the erosion problem [52]. It has been demonstrated that the TFI methods not only were able to reconstruct a larger region of the susceptibility maps, but also yielded improved susceptibility map estimation near the brain regions with large susceptibility differences such as the brain boundary and the interface between intracerebral hemorrhage (ICH) and surrounding tissue, compared to several Laplacian-based single-step QSM methods [52,53].

While all of these single-step methods yield superior results compared to those obtained from the multistep QSM methods (see Fig. 16.8 for example results), they are time-consuming iterative algorithms that rely on carefully chosen regularization parameters, limiting their practicality in clinical settings. Recently, several deep learning-based QSM methods have been proposed to make QSM become more practical by not only reducing computational workload through the use of non-iterative algorithms without requiring users to select regularization parameters at test time, but also providing

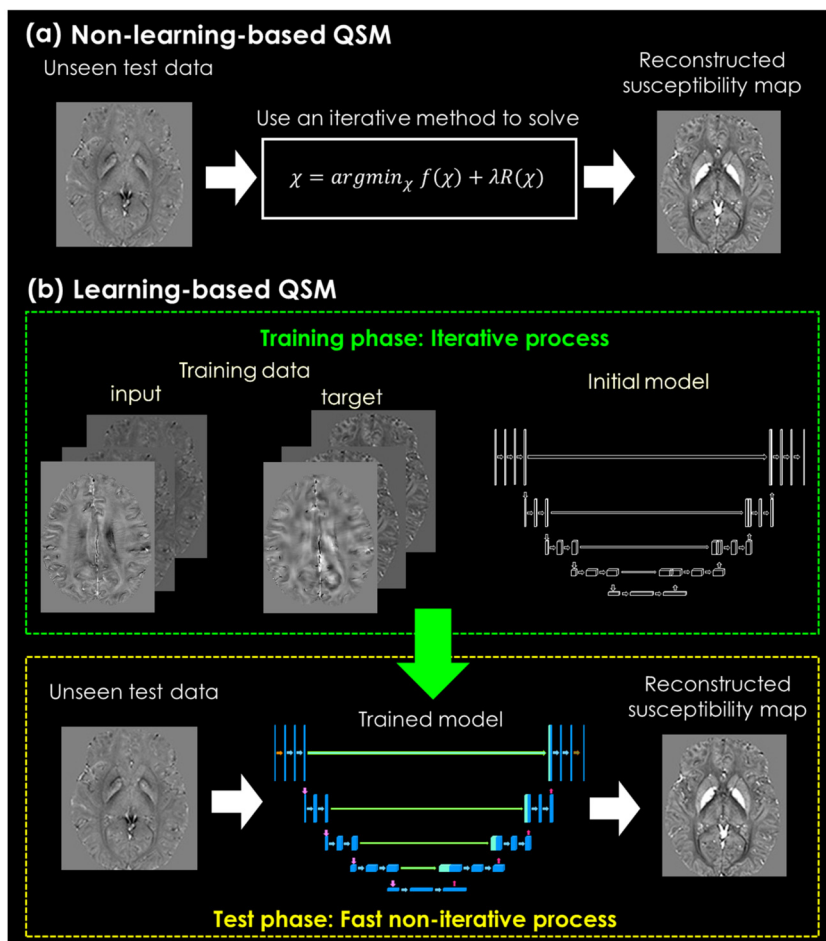
**FIGURE 16.8**

Comparisons between the TGV-regularized multistep QSM and TGV-regularized single-step QSM methods on (a) numerical brain phantom and (b) 7T *in vivo* brain data. Residual artifacts are clearly visible in the reconstructed susceptibility maps obtained from the multistep TGV method (indicated by the yellow (light gray in print version) arrows).

improved reconstruction quality [45,54,63–72,55,73–81,56–62]. These deep learning-based approaches often use high-quality COSMOS images acquired on multiple subjects as reference data.

A deep learning-based method (please see also Chapter 11) typically requires large amounts of data to train its model to perform the given tasks successfully, resulting in a training phase that is computationally expensive due to iterative optimization. Nevertheless, the test phase can be done much more quickly. The fast inference time of deep learning-based methods is an attractive characteristic for the practical aspect of QSM (Fig. 16.9). Developing deep learning-based QSM methods is thus a promising direction to accelerate QSM towards clinical applications.

With the availability of increasing amounts of QSM data, deep learning has rapidly gained success in the QSM community. In particular, deep learning has been used to improve each of the processing steps in QSM: phase unwrapping [54–56], background-field removal [57–59], and dipole inversion [45, 59,68–74,60–67]. Please refer to Table 1 of Ref. [75] for a comprehensive list of these deep learning-based QSM methods with their key features. The authors of [75] have also put together a list of the deep learning-based QSM implementations that are available online on <https://github.com/dlQSM/>.

**FIGURE 16.9**

(a) Non-learning-based QSM methods reconstruct the magnetic susceptibility map using an iterative algorithm. (b) Learning-based QSM methods usually consist of two phases: training and test phases. The training phase creates an initial model and iteratively updates the model using training data. The test phase reconstructs the susceptibility map using a single forward pass through the trained model. This non-iterative process in the test phase enables fast magnetic susceptibility reconstruction.

In addition to improving each of the QSM processing steps separately, deep learning-based methods that perform some of the processing steps simultaneously have been proposed [76–81]. Heber et al. have proposed a deep learning-based single-step QSM using a single fully convolutional network that is a concatenation of two U-shaped subnetworks: one performing the background field-removal step and the other performing the dipole inversion step [76]. The proposed method uses a simulated total phase map as its input and the corresponding susceptibility map in the region-of-interest as its target (or labels)

to train its two subnetworks to perform the two steps simultaneously. This is in contrast to the deep learning-based QSM method proposed in [59] that trains its two subnetworks separately. Liu and Kock have proposed deep learning-based single-step QSM methods that use a single 3D encoder–decoder network [77,78]. The proposed methods use the whole brain total-field map and brain mask as the inputs to the network and reconstruct both the underlying magnetic susceptibility map and the local field map simultaneously in [77], or reconstruct only the susceptibility map in [78]. Similarly, Gessner et al. explored the possibility of reconstructing the susceptibility map directly from the total field map and the mask defining the background regions using a single two-channel-input U-Net [79]. Kames et al. have proposed variants of variational network-based single-step QSM methods that estimate the magnetic susceptibility map from the total phase and an initial estimate of the susceptibility map [80]. Wei et al. have developed the deep learning-based *autoQSM* that reconstructs the magnetic-susceptibility map directly from the total phase map without requiring brain-volume extraction, bypassing the error-prone organ-masking process that frequently causes “erosion” problem [81].

Deep learning-based methods work especially well when training data and test data are drawn from the same underlying probability distribution. However, in most practical applications, the underlying probability distribution is unknown. Consequently, the success of many deep-learning models rely heavily on having access to large amounts of training data with enough variability to ensure that the model can be generalized to unseen test data. Unfortunately, it is costly to acquire extensive QSM data with enough variability. With the limited amounts of training data, the degraded performance of deep learning-based QSM methods can be observed when the unseen data characteristics differ from those of the training data [70]. It has been shown that QSMNet [61], when trained with data from only healthy volunteers, underestimated the susceptibility values in the hemorrhagic lesions of the patient test sets [70]. The different susceptibility dynamic range between the training data (-0.41 ppm– 0.61 ppm) and the test data (approximately -0.5 ppm– 1.40 ppm) led to the degraded performance of QSMNet. In order to mitigate this out-of-distribution problem, the authors augmented the training data by scaling their susceptibility values and demonstrated improved performance on the test set [70]. In addition to the susceptibility dynamic range mismatch, the resolution mismatch, different noise levels of the tissue phase, and different B_0 orientations also lead to degraded performance [75,79,82].

There are several approaches that can be adopted to mitigate this out-of-distribution problem in QSM, some of which have already been mentioned in the preceding paragraphs. Synthetic QSM datasets can be simulated with a wide range of conditions, such as different spatial resolutions, SNRs, B_0 orientations, and magnetic-susceptibility dynamic ranges. More data can also be generated by applying data augmentation to existing *in vivo* data such as scaling susceptibility values as done in [70], spatially shifting and rotating the data, and passing the estimated susceptibility map through a physics-based forward model. Moreover, widespread data-, model-, and code sharing within the QSM community would rapidly enable access to large QSM datasets and also better benchmarking of different QSM methods. Combining both synthetic and *in vivo* datasets with data augmentation techniques for training could also help with the generalization capability. Regarding the model training procedures, additional types of loss functions can be incorporated into the main objective function, such as the adversarial loss [63,83], perceptual/feature loss [84], or loss functions specifically designed to address susceptibility anisotropy, to complement the more commonly used Mean-Square-Error (MSE) and Mean-Absolute-Error (MAE) losses, which are pixel-based losses that are not designed for capturing perceptual characteristics. Furthermore, developing semi-supervised, self-supervised, or unsupervised learning-based methods [74,85] (see also Chapter 11) that typically require training data along with

fewer or none of the targets (labels) than commonly used supervised-learning-based QSM methods would also be a promising complementary approach toward bringing QSM to clinical applications.

16.6 Summary and outlook

QSM is a novel biomarker that has been linked to several physiological and pathological conditions. However, the required pre-processing and reconstruction steps may seem involved, and this chapter aimed to provide an overview and guideline to these numerous processing steps. In the following, the reconstruction pipeline will be briefly summarized and some recommendations for practical use will be given. We conclude this section with an outlook and information about ongoing communal efforts.

QSM reconstruction typically involves several pre-processing steps that convert the raw GRE data to a tissue field map. This pre-processing typically starts with the coil combination of multichannel data where vendor implementations of Roemer combination help estimate and remove phase offsets between coil channels. In the absence of such implementations, ESPIRiT can be used as a powerful alternative. The resulting coil-combined image is then brain masked to improve the efficiency of subsequent processing steps, and we recommend the use of FSL BET as a rapid and robust tool. As a next step, the image phase needs to be unwrapped to remove 2π phase jumps and obtain a global phase map. This processing step can be carried out in several different ways and established algorithms either rely on path-following or Laplacian techniques, which often result in comparable quality as recently demonstrated by an excellent review article [86]. As part of this contribution, we also provided source code and example data for Laplacian Unwrapping (see Tutorial 16.A.1) which should allow the interested readers to become familiar with this important processing step. Last, any unwanted background-phase contributions need to be removed to obtain a tissue phase map. In this chapter, we discussed two of the most widely used methods for background-phase removal, PDF and SHARP, but we would like to refer the interested readers to the following review article [29] for further information.

Next, the desired quantitative susceptibility is extracted from the tissue phase, and this step involves the solution of a complicated inverse problem. This is because the dipole kernel in k-space undersamples the frequency content of the desired susceptibility map and thus physically meaningful solutions can only be obtained through data oversampling or the use of prior information. The COSMOS algorithm oversamples the dipole inversion by jointly reconstructing data acquired in multiple orientations with respect to the main magnetic field, and this technique is often referred to as a gold standard due to its high-quality reconstructions. However, the need for reacquisitions in multiple orientations (at least three) and the associated long scan times limit its widespread clinical usage. In contrast, regularized algorithms enforce additional constraints, such as image smoothness to facilitate single-direction reconstructions, and MEDI is among the most widely used techniques. However, to trade-off accuracy/smoothing vs. streaking artifacts, regularization parameter(s) need to be well-chosen, and in Tutorial 16.A.2 we demonstrate the need for parameter tuning, which can be a tedious task. Moreover, regularized reconstructions often do not permit closed form solutions but require time-consuming iterative optimization that can be prohibitive for clinical usage.

Given these considerations, our overall recommendation for an easy-to-use, robust pipeline is to unwrap the images with Laplacian unwrapping (sum the unwrapped phase images across the echoes in a multi-echo acquisition), perform V-SHARP background-field removal, and continue with NDI dipole inversion. These steps have been implemented in Tutorials 16.A.1 and 16.A.2.

In this chapter, we also discussed recent developments in QSM. Single-step algorithms combine phase unwrapping, background-field removal, and dipole inversion into a single optimization problem. This was shown to reduce error propagation between the different processing steps thus yielding improved susceptibility-map estimation especially near regions with large susceptibility differences, such as the brain boundary and interfaces between intracerebral hemorrhage (ICH) and surrounding tissue. In Tutorial 16.A.3, we also provided example code that enables the interested readers to explore this novel field of research. However, while single-step methods generally yield superior image quality than classical multistep approaches, they entail an even more complicated reconstruction with increased computational footprint. In addition, regularization parameters still need to be chosen carefully to prevent oversmoothing, which poses yet another challenge for clinical adoption. Therefore, recent advances in deep-learning reconstruction may present a viable alternative to classical physics-based algorithms. These techniques have shown improvement across all QSM processing steps, offering improved image quality and faster reconstruction. Moreover, these methods no longer necessitate manual tuning of regularization parameters and thus offer greater practicability for clinical deployment. However, to achieve robust reconstructions, supervised deep-learning networks require training across large representative sets of training cases, and the availability of such datasets remains challenging and costly. Further, supervised deep-learning approaches are not guaranteed to generalize to unseen patient data and may thus result in undesirable image artifacts and distortions. These issues are often a consequence of limited training data and are increasingly tackled through synthetic/simulated data as well as data sharing.

To foster more widespread clinical usage of QSM, it will be crucial to resolve the remaining implementation challenges, and we believe that the use of deep learning could play an important role in this. In addition, more clinical studies will be needed to identify useful applications and increase the clinical utility of QSM. This may also increase the likelihood of commercial vendors to offer online scanner implementations and thus obviate the current need for offline data processing.

Ever since the advent of the Susceptibility Weighted Imaging (SWI) approach [10], there has been tremendous interest in mapping the phase and susceptibility distributions in the brain, with recent demonstrations in abdominal [6] and cardiac imaging [7]. The growth in the QSM community has been fostered by five ISMRM-sponsored workshops and two reconstruction challenges [87,88] to date. EMTP Hub (<https://www.emtphub.org/>) aims to bring QSM, as well as electrical properties-mapping communities together, and provide a common platform to share code, data, literature, and tutorials. We look forward to seeing these communal efforts on technical development to help deploy QSM in neuroscientific studies and clinical applications. For further information on the physics and applications of QSM, we refer the interested readers to excellent review articles in [2,89].

Appendix 16.A Tutorials

16.A.1 Phase pre-processing

16.A.1.1 *Provided materials*

- **main_phase_preprocessing.m** The main file that loads exemplar 3D-GRE data, and performs phase unwrapping and V-SHARP filtering.
- **imagesc3d.m** This function displays 3D data such as a magnetic-susceptibility map.
- **laplacian_unwrap.m** This function performs Laplacian unwrapping.

- **create_SMVkernel.m** This function creates spherical mean value (SMV) kernels.
- **msk.mat** Brain mask obtained from FSL BET.
- **prot.mat** Header information related to the 3D-GRE acquisition.
- **IMG.mat** Complex-valued 3D-GRE data that were acquired with five head orientations. The first orientation is processed in the main function.

16.A.1.2 Exercises

- (a) Display and compare V-SHARP reconstructions by varying the maximum SMV radius in *main_phase_preprocessing.m*.
- (b) V-SHARP result can be deconvolved with the response function of the largest SMV filter. This deconvolution benefits from additional regularization. Observe the effects of varying the regularization parameter on the final deconvolved tissue field map.

16.A.2 Dipole inversion

16.A.2.1 Provided materials

- **main_dipole_inversion.m** The main file loads the pre-processed tissue phase, GRE magnitude, and brain mask data. TKD and NDI are computed on single-orientation data, while COSMOS utilizes data from five head orientations.
- **imagesc3d.m** This function displays 3D data such as a magnetic-susceptibility map.
- **ndi_gradient_descent.m** This function performs the iterative NDI gradient-descent optimization
- **mag_dir1.mat** Magnitude image (single-orientation)
- **tissue_phase_dir1.mat** Tissue phase image (single-orientation)
- **mask_dir1.mat** Brain mask (single-orientation)
- **tissue_phase_all.mat** Tissue phase images from five head orientations (used for COSMOS only)
- **rotation_matrix_all.mat** Rotation matrices obtained from coregistration of multi-orientation tissue-phase images (used for COSMOS only)

16.A.2.2 Exercises

- (a) Display and compare the reconstructions of COSMOS, TKD, and NDI by running *main_dipole_inversion.m*.
- (b) COSMOS calculates the susceptibility map from multi-orientation data. Assess the effect of using data from only 2, 3, 4, and 5 head orientations.
- (c) TKD utilizes a k-space truncation parameter ϵ . Display and compare TKD for various values of ϵ .
- (d) NDI relies on a gradient-descent optimization, which is prone to overfitting when too many iterations are performed. Compare the image quality of NDI after 50, 200, and 1000 iterations. Besides early-stopping, overfitting can also be prevented by the use of Tikhonov regularization. Set $\lambda = 1e - 3$, and rerun NDI for a large number of iterations.

16.A.3 Total variation regularized single-step QSM (single-step TV)

16.A.3.1 Provided materials

- **SS_TV_QSM.m** An incomplete implementation of the Single-Step TV method proposed in [16,50].

- **main_single_step.m** The main file for parts (a)–(c) of this tutorial. It loads the relevant data, estimates the magnetic susceptibility map using the method implemented in *SS_TV_QSM.m*, and displays the results.
- **main_multi_step.m** The main file for part (d) of this tutorial. It loads the relevant numerical data, estimates the magnetic susceptibility map using two multistep QSM methods, and displays the results.
- **main_in_vivo.m** The main file for part (e) of this tutorial. It loads the relevant in vivo data, estimates the magnetic susceptibility map using Single-Step TV and two multistep QSM methods, and displays the results.
- **create_dipole_kernel.m** This function creates a dipole kernel in k-space.
- **create_SMVkernel.m** This function creates Spherical-Mean-Value (SMV) kernels.
- **imagesc3d.m** This function displays 3D data such as a magnetic-susceptibility map.
- **MS_L2_QSM.m** An implementation of the multistep QSM denoted as V-SHARP L2 in [16,50].
- **MS_TV_QSM.m** An implementation of the multistep QSM denoted as V-SHARP TV in [16,50].
- **Data/numerical_qsm_phantom.mat** The numerical brain phantom used for this tutorial.
- **Data/Wave_Caipi_7T.mat** The 7T in vivo brain data used for this tutorial.
- **poly_sensitivity_fit_coils.m** This function is used to remove a bias field from the magnitude images in part (e) of this tutorial.

16.A.3.2 Problem description

Conventional QSM methods that consist of multiple processing steps suffer from potential error propagation between the steps. In order to mitigate such errors, several single-step QSM methods have been proposed [16,40,46–53]. In this tutorial, we will implement the Single-Step TV method proposed in [16,50]. Specifically, the magnetic susceptibility map is estimated by solving the following regularized optimization problem:

$$\underset{\chi}{\text{minimize}} \frac{1}{2} \sum_i \left\| M_i F^{-1} H_i D F \chi - M_i F^{-1} H_i F \Psi(\phi) \right\|_2^2 + \alpha \|G\chi\|_1, \quad (16.T3.1)$$

where χ is the underlying magnetic-susceptibility map to be estimated from the acquired raw phase ϕ , Ψ is the Laplacian unwrapping operator divided by $2\pi * B_0 * \gamma * TE$, F is the discrete Fourier transform operator, D is the dipole kernel in k-space, M_i is a binary mask for the i^{th} reliable phase region, H_i is the Fourier transform of an SMV kernel for the i^{th} reliable phase region, $G = [G_x, G_y, G_z]^T$ is the gradient operator with G_x , G_y , and G_z being the gradient operators along the x-, y-, and z-axes, respectively. The first term of Eq. (16.T3.1), commonly called the data consistency/fidelity term, ensures that the reconstructed susceptibility map when passed through the forward model is consistent with the acquired data ϕ . The second term imposes prior information on the reconstructed susceptibility map. α is the regularization parameter balancing the effects of the first and second terms. Different α values result in solutions with different characteristics.

While Eq. (16.T3.1) can be solved directly using standard optimization algorithms, Chatnuntaweich et al. proposed to transform the problem into the following form:

$$\underset{\chi, z_1, z_2}{\text{minimize}} \frac{1}{2} \sum_i \left\| M_i F^{-1} z_{2,i} - M_i F^{-1} H_i F \Psi(\phi) \right\|_2^2 + \alpha \|z_1\|_1 \quad (16.T3.2)$$

subject to $G\chi = z_1$, and $H_i D F \chi = z_{2,i}$,

where z_1 and z_2 are auxiliary variables. It can then be shown that the problem in Eq. (16.T3.2) can be decomposed into simpler subproblems with the following closed-form updates [50]:

$$F\chi^{[k+1]} := \left(\mu_1 E^* E + \mu_2 D^* \sum_i H_i^* H_i D \right)^{-1} \left(\mu_1 E^* F \left(z_1^{[k]} - s_1^{[k]} \right) + \mu_2 D^* \sum_i H_i^* \left(z_{2,i}^{[k]} - s_{2,i}^{[k]} \right) \right) \quad (16.T3.3)$$

$$z_{2,i}^{[k+1]} = F \left(M_i^* M_i + \mu_2 I \right)^{-1} \left(M_i^* M_i F^{-1} H_i F \Psi(\phi) + \mu_2 F^{-1} \left(H_i D F \chi^{[k+1]} + s_{2,i}^{[k+1]} \right) \right) \quad (16.T3.4)$$

$$z_1^{[k+1]} = \left(\left| G\chi^{[k+1]} + s_1^{[k]} \right| - \frac{\alpha}{\mu_1}, 0 \right) \cdot \text{sign} \left(G\chi^{[k+1]} + s_1^{[k]} \right) \quad (16.T3.5)$$

$$s_1^{[k+1]} := s_1^{[k]} + G\chi^{[k+1]} - z_1^{[k+1]} \quad (16.T3.6)$$

$$s_{2,i}^{[k+1]} := s_{2,i}^{[k]} + H_i D F \chi^{[k+1]} - z_{2,i}^{[k+1]} \quad (16.T3.7)$$

where s_1 and s_2 are the scaled dual variables, μ_1 and μ_2 are the augmented Lagrangian parameters, and k in the square brackets is an iteration counter. An optimal solution to Eq. (16.T3.2) can be found by iterating Eqs. (16.T3.3)–(16.T3.7) until convergence.

16.A.4 Exercises

- Complete the implementation of the Single-Step TV method based on Eqs. (16.T3.3)–(16.T3.7) by modifying the parts marked as “**TO DO**” in *SS_TV_QSM.m*.
- Display the reconstructed magnetic-susceptibility maps generated by running *main_single_step.m*.
- Investigate the effects of α in Eq. (16.T3.2) on the reconstructed magnetic susceptibility map.
- Compare the reconstructed magnetic susceptibility map obtained from running *main_single_step.m* with your chosen α to several multistep QSM methods by running *main_multi_step.m*. Your Single-Step TV implementation should work better than the multistep QSM methods.
- In the previous parts, you have tested your implementation on the numerical brain phantom. In this part, you will test it on 7 Tesla in vivo data. Display the reconstructed magnetic-susceptibility maps generated by running *main_in_vivo.m*.

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